5 Sophie drives a distance of 513 kilometres on a motorway in France.
She pays 0.81 euros for every 10 kilometres she drives.

(a) Work out an estimate for the total amount that Sophie pays.

See worked solutions, page 2

..... euros

(3)

(b) Is your answer to part (a) an underestimate or an overestimate?
Give a reason for your answer.

.....

.....

.....

(1)

AI-generated worked solutions - may contain mistakes

AI-generated worked solutions

Written by an AI and may contain mistakes. Answers marked as not confidently solved or not checked need a careful look.

Question 5(a) (3 marks)

Sophie drives a distance of 513 kilometres on a motorway in France. She pays 0.81 euros for every 10 kilometres she drives. (a) Work out an estimate for the total amount that Sophie pays.

Not checked by calculation: look over this one. Two of three attempts at this task reached this answer (supporting evidence, not a check by calculation).

From the question: distance = 513 km; cost per 10 km = 0.81 euros.

Estimate the distance by rounding to a multiple of 10.

$513 \approx 510$

Calculate the cost for the estimated distance using the rate.

$$\frac{510}{10} \times 0.81 = 51 \times 0.81 = 41.31$$

Answer: 41.31euros

Yes, the answer is an underestimate because 510 km is less than 513 km. The method involves rounding the distance to the nearest convenient number (510) and then multiplying by the cost per unit distance.

Question 5(b) (1 mark)

Sophie drives a distance of 513 kilometres on a motorway in France. She pays 0.81 euros for every 10 kilometres she drives. (b) Is your answer to part (a) an underestimate or an overestimate? Give a reason for your answer.

AI-generated worked solutions (continued)

Partly worked: not finished or not consistent, so look it over. The answer is not stated clearly in words, with its reason. Look it over before relying on it.

From the question: distance = 513 km; cost per 10 km = 0.81 euros.

Round the distance to the nearest 10 km to get a quick estimate.

$513 \approx 510$

Calculate the estimated cost using the rounded distance.

$$\frac{510}{10} \times 0.81 = 51 \times 0.81 = 41.31$$

Answer: overestimate

No, it is an overestimate because the rounded distance of 510 km is less than the actual distance of 513 km, so the calculated cost is lower than the true cost. Rounding 513 down to 510 gives a smaller number of 10-km blocks, leading to a lower estimated total. Since we rounded down, the estimate is an underestimate.